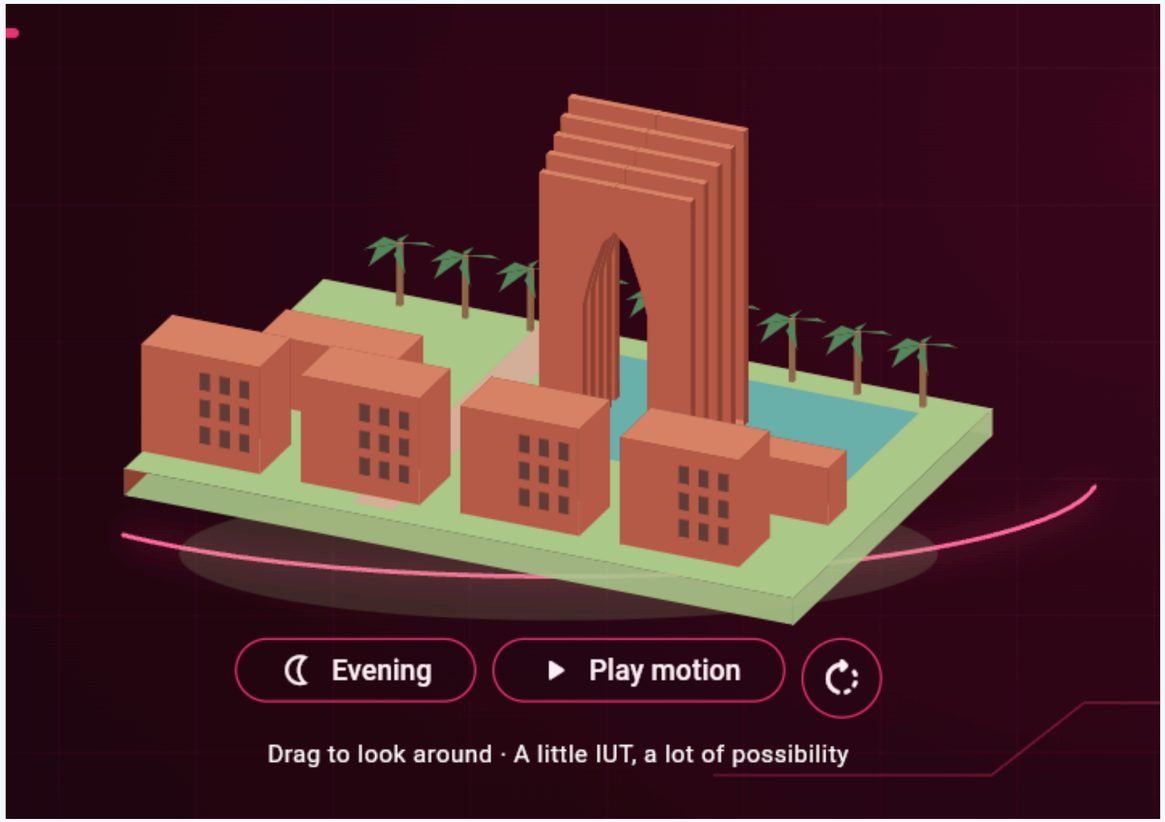
UniConnect

Course Title: Software Project Lab II
Course Code: SWE 4404

University Social Networking Platform

FINAL PRESENTATION



TEAM-4

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PROJECT FEATURES

1. **Authentication & Authorization**
2. **Project Teammate Finder**
3. **Peer-to-peer Mentorship**
4. **Career Board**
5. **Events & Announcements**
6. **One-on-One Chat**

PROJECT STRUCTURE

REPOSITORY OVERVIEW

```
SPL-II_UniConnect/
├─ code/
│  ├─ backend/
│  │  ├─ src/main/java/com/uniconnect/
│  │  │  ├─ shared/
│  │  │  ├─ authentication/  profile/
│  │  │  ├─ discovery/  connection/  chat/
│  │  │  ├─ club/  project/  career/  mentorship/
│  │  │  └─ notification/  administration/  reporting/
│  │  └─ resources/db/migration/
│  └─ frontend/lib/
│     ├─ core/
│     └─ features/
├─ docs/
└─ .github/workflows/
```

MVC Architecture

- **Model:** Services, domain logic, and database operations.
- **View:** Flutter user interface.
- **Controller:** Handles REST/WebSocket requests.

Layer patterns

Backend

Controller → DTO → Service → Model → Repository

Flutter

View → State → Model → Data

COMMUNICATION BETWEEN COMPONENTS

Flutter UI
Views & state

→

REST API
JSON / DTOs

→

Spring Boot
Controllers & services

→

JPA / Hibernate
Repositories

→

PostgreSQL
Persistent data

REAL-TIME

WebSocket Chat & live notifications

EXTERNAL SERVICE

SMTP email Verification & password reset

ONE COMPLETE FEATURE REQUEST

Example: Registration and numeric email verification

01 **Flutter**

Collect sign-up details

02 **REST / JSON**

POST /api/v1/auth/register

03 **Controller**

Validate and delegate

04 **Service**

Enforce rules; hash password

05 **Database**

Save as PENDING_VERIFICATION

06 **Verification**

Email code → ACTIVE

DIVISION OF CLASSES & METHODS

Layer	Reason for separation
Controller	HTTP validation and delegation
Service	Business rules and transactions
Repository	Database queries
DTO / Mapper	Request and response conversion
Domain	State transitions and invariants

LOC metric	Measured
Maximum class	757
Average class	54.88
Maximum method	287
Average method	12.35

Single responsibility • clear dependency direction • testable services

Thank You

TEAM 4 • SWE 4404

github.com/aymannondiny/SPL-II_UniConnect

APPENDIX I

Registration request

01 **Flutter form and request**

Full name, IUT email, password, confirmation and Student/Alumni selection

02 **POST /api/v1/auth/register**

JSON: fullName, email, password, platformRole

03 **AuthenticationController.register()**

@Valid RegisterRequest checks fields, then delegates to RegistrationService

04 **RegistrationService.register()**

Normalize email, enforce role/domain and password rules, reject duplicates, BCrypt hash

APPENDIX II

Email verification and login

05 **Persist and send**

Save User as PENDING_VERIFICATION and a salted code digest. Send six-digit code by SMTP.

06 **201 Created returns RegisterResponse**

userId, fullName, email, platformRole, accountStatus. Flutter opens code entry.

07 **POST /api/v1/auth/verify-email**

Body: email + code. Check expiry and attempts, consume code and activate account. Return 204.

08 **Log in separately**

Flutter opens Login. A successful login creates the authenticated session.